

# NOYAL SAM D

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## PROFESSIONAL SUMMARY

AI/ML & Data Science professional specializing in Generative AI, machine learning, and NLP. Proficient in LangChain, LangGraph, RAG, PyTorch, TensorFlow, and Scikit-learn. Skilled in building LLM-powered applications, end-to-end ML pipelines, and AI agents. Strong in feature engineering, model evaluation, and production-ready model deployment.

## SKILLS

**Programming:** Python, SQL

**ML & Deep Learning:** Scikit-learn, XGBoost, TensorFlow, Keras, PyTorch, Model Training, Evaluation & Deployment

**Generative AI & LLMs:** LangChain, LangGraph, LCEL, Prompt Engineering, RAG, AI Agents, Hugging Face, Embedding Models, FAISS, MCP, OpenAI API, Groq API, Google Gemini API, Ollama, Fine Tuning

**NLP & Computer Vision:** NLTK, spaCy, NER, Tokenization, Text Preprocessing, Text Classification, CNN, Image Classification

**Data Science & Analysis:** Data Preprocessing, EDA, Feature Engineering, Statistical Analysis, Data Pipelines

**Big Data :** PySpark, Hadoop, Databricks

**Analysis & Visualization:** NumPy, Pandas, Matplotlib, Seaborn, Power BI, Excel

**Databases :** MySQL, Oracle SQL

**Tools, Cloud & DevOps:** AWS (EC2, S3), Docker, CI/CD, Git, GitHub, Flask, REST API, Streamlit

## EXPERIENCE

**AI/ML Engineer Intern | PySpiders, Bengaluru**

**June 2025 – Present**

- Built and deployed NLP-based RAG chatbot systems using LangChain, FAISS, embeddings, and LLMs, enabling semantic search and intelligent question answering over 500+ pages of document repositories.
- Engineered end-to-end machine learning data pipelines including data preprocessing, feature engineering, model training, and evaluation, improving model F1-score by 18% through iterative optimization.
- Developed Generative AI applications using Prompt Engineering, RAG architectures, vector databases, and LLM frameworks, reducing manual research effort by 40% through intelligent automation.

**Data Science Intern | EX Life Solution, Thiruvananthapuram**

**June 2024 – August 2024**

- Built end-to-end ML data pipelines in Python on 1M+ row datasets, reducing data preparation time by 35% through reusable preprocessing modules for feature engineering and normalization using Pandas, NumPy, and Scikit-learn.
- Performed statistical analysis and tuned regression, classification, and ensemble models (Random Forest, XGBoost) using cross-validation and hyperparameter optimization, achieving 89% accuracy on held-out test sets.

## PROJECTS

**Plant Disease Detection System Using Deep Learning | [GitHub](#)**

- Developed a Computer Vision-based multi-plant leaf disease detection system using a custom 3-layer CNN (Conv2D, ReLU, MaxPooling) trained on 30,000+ images with Adam optimizer, achieving 94.78% accuracy across 8 disease categories.
- Built and deployed a Flask-based REST API with multi-route architecture enabling real-time model inference with preprocessing techniques such as Gaussian blur and normalization.

**Agentic RAG-Based Research Assistant | [GitHub](#)**

- Built an agentic RAG system using LangGraph ReAct Agent and Llama 3 (Groq) with FAISS vector store, Hugging Face embeddings, and Tesseract OCR to process PDFs and images for semantic search and question answering.
- Developed an interactive Gradio UI integrating LangChain-based document retrieval and real-time web search, enabling end-to-end agentic reasoning over private and live data.

**Code Error Detection and Remediation Engine | [GitHub](#)**

- Developed an AI-powered two-stage LangChain pipeline using Gemini 2.5 Flash to detect syntax, runtime, security, and logical errors across 5+ programming languages with severity ratings and auto-corrected code output.

**AI-Powered Adaptive Learning Assistant | [GitHub](#)**

- Developed an AI-powered personalized learning assistant using LangChain and Gemini that explains concepts, answers subject-specific queries, and delivers adaptive learning support through a conversational interface.

**Personalized Career Roadmap Generator | [GitHub](#)**

- Built a RAG-powered career roadmap generator using LangChain LCEL, FAISS, and Gemini 2.5 Flash with dynamic timeline calculation based on skill gap analysis, deployed as a Flask web application supporting 26 career profiles.

## EDUCATION

**B.Tech – Information Technology**

**7.69 CGPA | 2021 – 2025**

St. Xavier's Catholic College of Engineering, Nagercoil